Chapter 4 Practice Quiz

Name: _____ Date: _____

Use g = 10 N/kg = 10 $\frac{\text{m}}{\text{s}^2}$. Numeric answers need units. For calculations, show the formula, the substitution, and the answer.

1. A cart moves to the right and is speeding up. Which way does the net force on it point?

(A) To the right
(B) To the left
(C) The net force is zero
(D) Can't tell without knowing its mass

2. A cart moves to the left. The net force on it points to the right. What happens to the cart?

(A) It speeds up.
(B) It moves at constant speed.
(C) It instantly moves to the right.
(D) It slows down.

3. Cart A and Cart B have the same mass. The net force on A is 4 N and the net force on B is 12 N. How do their accelerations compare?

(A) B's acceleration is 1/3 of A's.
(B) They are the same.
(C) B's acceleration is 3 times A's.
(D) B's acceleration is 9 times A's.

4. Which statement about mass and weight is correct?

(A) Mass is measured in kg and weight is measured in N.
(B) Mass is measured in N and weight is measured in kg.
(C) Both are measured in kg.
(D) Both are measured in N.

5. A ball is thrown straight up. While it is on its way up (before the top), which is true? Ignore air resistance.

(A) It slows down because its acceleration points upward.
(B) It speeds up because its acceleration points downward.
(C) It moves at constant speed because its acceleration is zero.
(D) It slows down because its acceleration points downward.

6. A car is moving east and slowing down. Which statement is true?

(A) Its acceleration points east.
(B) Its net force points west.
(C) Its net force is zero.
(D) Its velocity points west.

7. Which two quantities always point in the same direction?

(A) Net force and acceleration
(B) Velocity and acceleration
(C) Velocity and net force
(D) Position and net force

8. A box sits at rest on the floor. You push it with 30 N and it does not move. The friction force on the box is:

(A) 0 N
(B) Less than 30 N
(C) Exactly 30 N
(D) More than 30 N

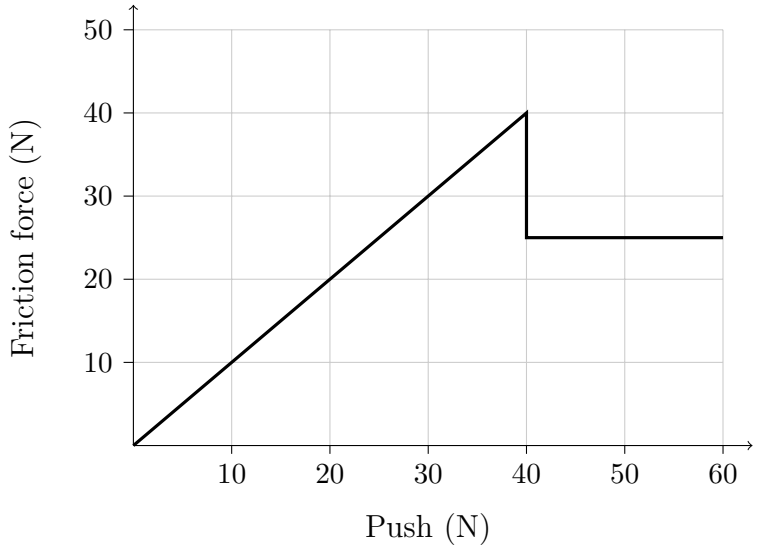
9. A heavy box is hard to start moving but easier to keep moving. This is because:

(A) Kinetic friction is greater than static friction.
(B) The maximum static friction is usually greater than the kinetic friction.
(C) There is no friction once the box moves.
(D) The box gets lighter when it moves.

10. Which change would increase the kinetic friction on a sliding crate?

(A) Sliding it faster
(B) Turning it so a smaller face is on the floor
(C) None of these. Friction can't change.
(D) Loading more mass into the crate

Use this graph for #11-13. It shows the friction force on a box as you push harder and harder.



11. What is the largest static friction force on the box?

(A) 25 N
(B) 30 N
(C) 35 N
(D) 40 N

12. You push the box with 30 N. The friction force on it is:

(A) 0 N
(B) 30 N
(C) 40 N
(D) 60 N

13. A 10 kg box is pushed with 55 N and slides across the floor.

(a) Use the graph to find the friction force on it.

(b) Find the net force.

(c) Find the acceleration.

Air resistance is proportional to the frontal area and to the square of the speed. Use this for #14, #15, and #28.

14. A car's speed triples. The air resistance on it becomes about:

(A) 3 times as big
(B) 6 times as big
(C) 9 times as big
(D) 27 times as big

15. A skydiver spreads out so that her frontal area doubles. At the same speed, the air resistance on her becomes about:

(A) The same
(B) 2 times as big
(C) 4 times as big
(D) 8 times as big

16. A skydiver falls at a constant speed (terminal velocity). Compared with her weight, the upward air resistance on her is:

(A) Zero
(B) Less than her weight
(C) Equal to her weight
(D) Greater than her weight

17. A rock is taken from Earth to the Moon. Its mass and its weight:

(A) Mass stays the same; weight is smaller.
(B) Both stay the same.
(C) Mass is smaller; weight stays the same.
(D) Both are smaller.

18. The value of g is 1.6 N/kg on the Moon, 10 N/kg on Earth, and 25 N/kg on Jupiter. Which list shows the weight of the same rock on each world, from least to greatest?

(A) Earth, Moon, Jupiter
(B) Jupiter, Earth, Moon
(C) Moon, Earth, Jupiter
(D) All the same

19. A student reports, "The net force on the cart was 15 kg." What is the problem?

(A) Nothing. It is correct.
(B) Force should be in m/s^2.
(C) Force should be in m/s.
(D) Force should be in N.

20. Which quantity is matched with the correct unit?

(A) Force: kg
(B) Mass: kg
(C) Acceleration: m/s
(D) Velocity: m/s^2

21. An object has a mass of 150 g. What is this in kg?

(A) 0.15 kg
(B) 1.5 kg
(C) 15 kg
(D) 150 kg

22. A net force of 18 N gives a cart an acceleration of 3 m/s^2. Find the mass of the cart.

23. A 500 g cart accelerates at 6 m/s^2. Find the net force on it (in N).

24. A 1000 kg car speeds up from 0 to 20 m/s in 10 s.

(a) Find its acceleration.

(b) Find the net force on the car.

25. A 40 kg student. The value of g is 10 N/kg on Earth and 1.6 N/kg on the Moon.

(a) Find her weight on Earth.

(b) Find her weight on the Moon.

(c) What is her mass on the Moon?

26. A 4 kg box moves to the right. A 25 N push acts to the right and 5 N of friction acts to the left.

(a) Find the net force (size and direction).

(b) Find the acceleration (size and direction).

27. A 2 kg ball is falling. At one moment, the air resistance on it is 8 N.

(a) Find the ball's weight.

(b) Find the net force on the ball.

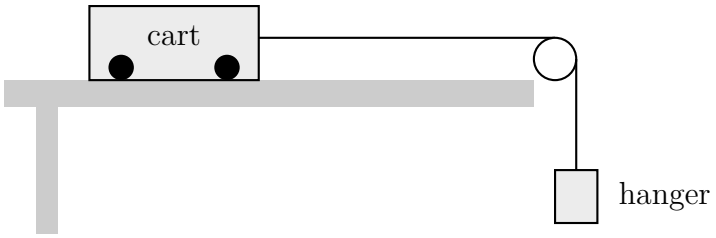
(c) Find the ball's acceleration at that moment.

28. At 10 m/s, the air resistance on a cyclist is 20 N. Remember that air resistance is proportional to the square of the speed. How big is the air resistance at 30 m/s?

29. A 3 kg rock is dropped on Mars, where g = 3.7 N/kg. There is no air resistance.

(a) Find the rock's weight.

(b) Find the rock's acceleration.



30. Cart = 1.4 kg and hanger = 0.6 kg.

(a) What is the total mass being accelerated?

(b) What is the pulling force?

(c) What is the acceleration?

31. Now the hanger is 0.4 kg and the system accelerates at 2 m/s^2. The mass of the cart is not known.

(a) What is the pulling force?

(b) What is the total mass being accelerated?

(c) What is the mass of the cart?